ICE: A Local-First Conversational Memory System and a Longitudinal Evaluation Protocol

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Abstract

Conversational assistants forget everything between sessions, forcing users to re-establish context continually. We present the Infinite Context Engine (ICE), a local-first conversational memory system that sits between the user and any OpenAI-compatible model and maintains persistent memory across arbitrarily long histories: episodic turns, a temporally-versioned knowledge graph, procedural patterns, and documents, retrieved through weighted rank fusion under a per-query token budget. Evaluating such a system is itself an open problem, because static benchmarks cannot capture memory that accumulates, decays, and is revised over months. We therefore introduce LSREP, a Longitudinal State-Replay Evaluation Protocol that replays real, self-authored conversations (1,985 turns; 1,211 probes at 50 checkpoints), reconstructs the full memory state at each checkpoint, and scores answers against an evolving ground truth. Under LSREP, ICE matches a strong vector-RAG baseline on answer quality (paired $\Delta = +0.00$, 95% CI $[-0.07, +0.07]$) while injecting 32% fewer fragments, and wins a materially larger share of blind head-to-head comparisons (30.6% vs. 21.2%, non-overlapping intervals); on an independently hand-scored slice the margin is $+1.21$ $[+0.81, +1.61]$. ICE's retrieved fragments correlate *positively* with answer quality ($r = +0.19$) where the baseline's correlate *negatively* ($r = -0.02$). Under extreme per-turn context density the unbudgeted baseline collapses (94.2% failure) while the budgeted system holds (mean 4.33). A cumulative ablation isolates rank fusion as the mechanism that makes multi-signal retrieval safe: an unfused lexical signal is harmful (-0.74, $[-1.14, -0.36]$) and fusion recovers it ($+0.82$, $[+0.39, +1.24]$). We pair these results with a component-level fidelity audit that separates three things systems evaluations routinely conflate: one component that was genuinely defective (a procedural-retrieval binding bug), several that this benchmark never exercised (document retrieval, batch summarisation, temporal and cross-conversation queries), and the mechanisms that demonstrably carried the result—the token budget, rank fusion, decay-weighted episodic retrieval, and post-fusion curation. Separating "broken" from "untested" is, we argue, a methodological requirement for evaluating any multi-component memory system, and the evidence here supports one design principle: memory is curation, not collection.

1 Introduction

A language model holds a conversation only for as long as its context window holds the conversation; everything older is silently gone, and the interface gives no warning that it happened. We call the resulting failure *context collapse.* Its clearest form occurs during creative or narrative work and in long technical discussions. The model usually retains the general idea of what is going on, but forgets specificity: the exact rule that was set, a particular character, a precise decision made yesterday, the dependency-injection pattern established three months ago. The conversation continues, but the model has quietly lost the thread. The workarounds—exporting chats, pasting into a fresh context, re-explaining everything for the third time—survive the first few exchanges of a new session and then degrade, because the assistant that started coherent is, by turn fifteen, an assistant that has lost the new context too.

This is not a minor inconvenience. For a person doing sustained work—building a software system over months, constructing an original fictional universe across hundreds of sessions, planning a multi-year academic trajectory—the lack of persistent, intelligent, user-controlled memory is the single largest bottleneck between the human and the model working as genuine collaborators. The systems designed to address it either cap at a few hundred lines of static instructions, go stale, or live in someone else's cloud with no user control over what is remembered and what is forgotten.

The Infinite Context Engine (ICE) is a local-first personal memory middleware for conversational interfaces. It sits between the user and their language models—classifying every prompt, deciding which memories to surface, routing to a model, managing the accumulation of knowledge over time, and injecting context into every call. ICE is infrastructure for the thinking layer: for sessions where the assistant needs to remember that two names refer to the same entity across different eras, that a database schema is partitioned a particular way, or that an architectural decision was made last Tuesday and its alternatives already ruled out.

A crucial architectural assumption is that the language model is a *stateless reasoning engine*. ICE does not embed memory into weights through fine-tuning; it externalises memory entirely, storing, maintaining, and retrieving context as an independent system surrounding a model that processes whatever context it is given and forgets it immediately. This separation lets models be swapped or upgraded without losing memory, and makes the store transparent, inspectable, and user-controllable. A vector-RAG baseline shares the superficial property of feeding external context to a stateless model, but is not *designed* to reconstruct a structured, evolving cognitive state: it stores raw chunks, with no knowledge graph, behavioural-pattern inference, or decay-and-reinforcement lifecycle.

Evaluating a system like this is a second, separable problem. A memory system's defining behaviours—accumulation, decay, revision of superseded beliefs—are invisible to a benchmark that asks questions once, at one moment, against a fixed ground truth. What is needed is a protocol that can stand a system at many points along a real conversational history and ask what it knows at each of them.

This paper makes three contributions.

1. **A local-first conversational memory system** (ICE): a multi-store architecture—episodic memory, a temporally-versioned knowledge graph, procedural memory, and document memory—with intent-driven retrieval routing, access-weighted decay, weighted rank fusion, and a dynamic per-query token budget. It runs on a single consumer GPU and is released in full.

2. **LSREP**, a Longitudinal State-Replay Evaluation Protocol. It replays months of real conversational history through a memory system, reconstructs the complete memory state at each of many temporal checkpoints, and scores answers against an *evolving* ground truth. LSREP is independent of ICE and is, we believe, the paper's most reusable contribution.

3. **A longitudinal evaluation with an explicit fidelity audit.** ICE matches a strong vector-RAG baseline on answer quality while injecting 32% fewer fragments and winning substantially more blind head-to-head comparisons; under a context-density stress test where the unbudgeted baseline fails on 94.2% of probes, ICE holds at a mean score of 4.33; and a cumulative ablation shows that an unfused lexical signal is harmful while rank fusion is corrective. We report alongside these results a component-level audit of which mechanisms were actually live, distinguishing one defective component from several the benchmark never exercised and from the mechanisms that carried the result (Sections 6.4 and 8.1).

The third contribution is also a methodological argument. Multi-component retrieval systems are routinely evaluated by ablation, and a near-zero delta is routinely read as "this component does not help." That inference is only valid if the component ran. We found by post-hoc audit that it sometimes had not—and that the fix is cheap: assert per-component contribution inside the harness, and report a contribution table beside the results. We report our own audit in full, including where it constrains what we may claim.

2 Related Work

We organise prior work into four areas and are deliberately specific about the gaps ICE attempts to address.

2.1 Stateful Memory Architectures for Dialogue

Bounded context length has driven a family of external memory hierarchies. **MemGPT** (Packer et al., 2023) treats the context window as primary memory backed by unbounded external storage, paging between tiers to create the illusion of unbounded context. It is effective on short-to-medium interactions and document analysis but degrades in extended multi-session dialogue: retrieval is flat and dense, it struggles to distinguish semantically identical but contextually distinct entities over time, and it has neither a structured graph tracking entity evolution nor intent-driven routing.

mem0 (Chhikara et al., 2025) targets the latency and token cost of paging with a lightweight production memory layer: a single-pass, append-only pipeline distils conversations into atomic facts, and a multi-signal retriever fuses embeddings, BM25, and entity linking, performing strongly on LoCoMo and LongMemEval at low overhead. The cost is cognitive depth: flat key-value facts are not versioned, so contradictory or evolving facts are never systematically reconciled, and there is neither procedural memory nor a principled decay and archival mechanism.

MemoryBank (Zhong et al., 2024) integrates an Ebbinghaus-style forgetting curve so every item carries a continuously updated strength score under time-aware decay and reinforcement, sustaining persona invariance across multi-day dialogue. It remains constrained by its retrieval substrate: no graph for associative multi-hop reasoning, and no conversation-scoped clustering. **SCMoE** (Shi et al., 2024a) attacks the cost of dense retrieval differently, contrasting strong and weak expert activations at inference; it optimises routing of internal representations rather than external memory, and is untested on the large, shifting per-turn documents that characterise persistent agentic memory.

Across these systems one trend recurs: static fact storage is conflated with dynamic working memory.

2.2 Knowledge Graphs as Dialogue Substrates

To overcome the relational blindness of flat vector stores, several systems map text into graphs. **GraphRAG** (Edge et al., 2024) answers global sense-making queries over private corpora by extracting an entity graph with a language model, partitioning it by community detection, and pre-generating hierarchical community summaries—a substantial gain in answer depth on query-focused summarisation. It assumes a temporally *static* corpus: all ingested facts are equally current, and there is no versioning for an entity whose properties or relationships mutate across sessions. **KGP** (Wang et al., 2024) builds a graph whose nodes are passages and whose edges are semantic or lexical similarities, then deploys a traversal agent as a local navigator for multi-document question answering; **Think-on-Graph** (Sun et al., 2024) couples the model tightly to the graph, running iterative beam search over reasoning paths and pruning irrelevant branches.

Both demonstrate the value of structural retrieval, but are tailored to single-shot question answering over static knowledge bases, treating the graph as ground-truth world knowledge. They contain no mechanism for narrative continuity, belief decay, or contradiction resolution—exactly what an evolving conversation demands. ICE borrows the topology and repurposes it for a continuously mutating conversational landscape.

2.3 Retrieval-Augmented Generation

RAG (Lewis et al., 2020) formalised augmenting parametric knowledge with a dense index over a static corpus. Conversational history is not a static corpus: it grows organically, facts are updated or contradicted, and context shifts. Subsequent work optimised the pipeline—**REALM** (Guu et al., 2020) folds a latent retriever into pre-training; **Fusion-in-Decoder** (Izacard and Grave, 2021) encodes passages independently before joint decoder attention, aggregating evidence over up to a hundred passages; **REPLUG** (Shi et al., 2024b) tunes the retriever against a frozen black-box model using perplexity signals; **CRAG** (Yan et al., 2024) adds a lightweight evaluator that scores retrieved documents and triggers a decompose-then-recompose filter. All four remain structurally confined to static knowledge bases, with no mechanism for a personal history in which preferences, constraints, and state of mind shift iteratively. ICE replaces the static-corpus assumption with a mutable, chronologically aware middleware.

2.4 Algorithmic Primitives

ICE synthesises several established primitives. **BM25** (Robertson and Zaragoza, 2009) scores relevance from a non-linear term-frequency function, inverse document frequency, and length normalisation; dense embeddings capture conceptual similarity but fail on exact-keyword matching, so BM25 provides lexical recall as a second leg. **Reciprocal Rank Fusion** (Cormack et al., 2009) combines rankings by the inverse of rank position across independent systems, requiring no score normalisation and preventing any one retriever from dominating; RRF is ICE's fusion backbone, and our ablation finds it *corrective*—it rescues a multi-signal retriever from the noise an unfused lexical leg injects (Section 6.3) rather than lifting quality above a strong single-leg baseline. **HyDE** (Gao et al., 2023) addresses vocabulary mismatch by embedding a generated hypothetical document instead of the query; ICE implements it as an optional gated rewrite, but it was not isolated in either experiment (Section 8.1), so we make no empirical claim about it.

2.5 The Gap

The gap is not any single missing feature; it is the combination of temporal awareness, structured entity memory, intent-driven routing, and a per-query token budget, evaluated over a history long enough for accumulation and revision to matter. Systems supplying one of these still fail on the case that motivated ICE: a long-running project in which entities evolve, intentions shift, and conversation history must be *filtered* as much as retrieved. The evaluation literature has a matching gap—LoCoMo and LongMemEval measure recall over fixed histories, not a memory state that ages between questions—which Section 5.2 addresses.

3 Datasets

The evaluation uses four long-form conversational datasets, all authored by a single user over periods ranging from months to years. We use anonymised descriptive labels rather than the internal names. table 1 summarises them.

Each was selected on three criteria: length well beyond typical chat interactions (251–1,119 turns), so information becomes separated by hundreds of turns; a distinct memory domain, since the creative datasets stress narrative continuity and entity consistency while the technical and academic ones stress design rationale and decision history; and dense memory structure—recurring entities, evolving relationships, procedural patterns, long-range dependencies. These conversations are demanding in ways synthetic benchmarks are not, but they represent one author, not a population; Section 8 returns to this.

Table 1: Four evaluation datasets (anonymised). All are single-user, self-authored. "Stress dimension" indicates which aspect of the architecture each dataset most heavily exercises.

Label	Description	Turns	Ckpts.	Stress dimension
A (Creative Writing)	Collaborative fan-fiction writing session	290	10	Narrative continuity, character evolution
B (Long-Form Creative)	Fantasy world-building and story-planning	1,119	20	Temporal-horizon retrieval, entity evolution
C (Technical Planning)	Design and development of ICE itself	325	8	Information density, 8K+ token turns
D (Academic Planning)	Graduate school and career planning	251	12	Mixed personal/technical, decision tracking
Total		**1,985**	**50**	

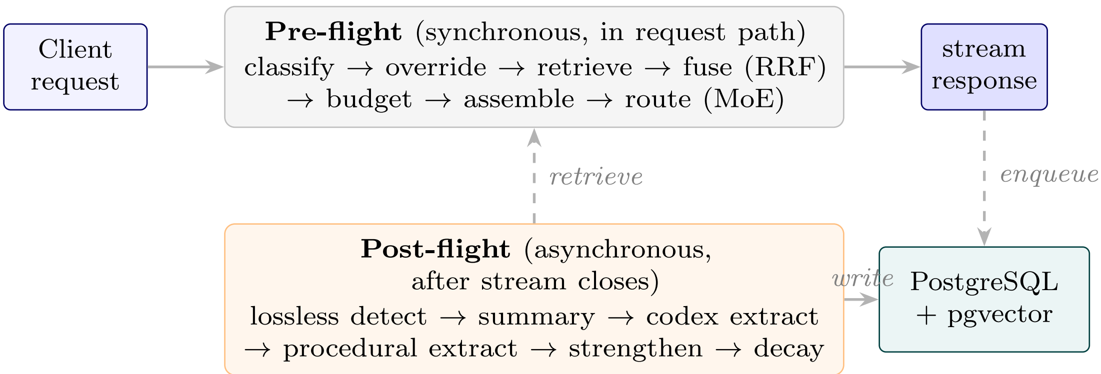

Figure 1: System overview. Pre-flight runs synchronously in the request path; post-flight runs asynchronously after the stream closes and feeds the next turn's retrieval.

4 System Architecture

ICE is an OpenAI-compatible memory middleware between a conversational client and a pool of locally-served models. It is not a model: every request addressed to the synthetic model name `ice-proxy` is intercepted by a proxy that classifies the turn, retrieves context from four long-lived memory stores, assembles a cache-friendly prompt, routes the request to a per-turn specialist model, streams the response back, and then dispatches background workers that extract, decay, cluster, and consolidate the new turn into long-term memory. Any downstream model therefore operates against an effectively unbounded personalised context, selectively rebuilt on every turn. This section describes the mechanisms the evaluation exercises; Appendix E carries the implementation detail (classification engine, Codex internals, prompt assembly, worker cluster, operational infrastructure), and the archived technical report in the repository describes the system exactly as evaluated.

4.1 Request Lifecycle

Each turn traverses a **pre-flight** phase (synchronous, in the request path) and a **post-flight** phase (asynchronous, after the stream closes).

In pre-flight, the user message is classified into topic tags, intent tags, and a single context-reliance label (Zero_Shot, Long_Term_Memory, Real_Time_Search) by a two-stage cascade: a rule-based pre-classifier resolves obvious cases in microseconds, and on a miss a small multi-layer perceptron over a frozen `Qwen3-Embedding-0.6B` encoder resolves the rest with sub-millisecond CPU inference. Hard override rules then coerce the label—most consequentially, a conservative rule upgrading Zero_Shot to Long_Term_Memory when a conversation exceeds ten turns or confidence falls below 0.95, so retrieval runs for almost every turn in a long conversation; Section 8 discusses what that costs. A Hybrid Retrieval Orchestrator then runs the retrieval legs, fuses them with weighted Reciprocal Rank Fusion, and post-processes the fused list. A Prompt Assembler concatenates the surviving fragments with persistent memory slots, recent turns, and the live message under a stable prefix, and a Mixture-of-Experts router selects a locally-served model with per-conversation stickiness.

In post-flight, an evaluator runs lossless detection—the guiding principle in the code is that *memory is earned*: a turn is stored verbatim only if dense enough, and is otherwise summarised—and dispatches idempotent extraction tasks for the knowledge graph and for procedural patterns. Scheduled workers (decay, clustering, reflection, batch summarisation, sentinel monitoring, and a weekly classifier fine-tune) maintain the stores over time; table 18 in the appendix lists them.

4.2 Memory Stores

ICE maintains four stores plus two structured overlays, all in one PostgreSQL database with the pgvector extension; every vector column is 384-dimensional because a single embedder is shared throughout.

Table 2: The four memory stores and the overlays. Each store is queried by one or more retrieval legs. Appendix E gives schemas and write paths.

Store	Purpose	Retrieval method
Episodic	Every conversational turn: raw text, summary, lossless flag, decay score, access count, embedding	BM25 leg; vector leg with decay weighting; session diversification
Codex	Temporally-versioned semantic graph: entities with aliases and properties, typed edges carrying `valid_from`/`valid_until`, append-only event log, snapshots	Graph traversal (BFS depth 3 over edges where `valid_until IS NULL`); enumeration fallback
Procedural	Recurring behavioural patterns: trigger conditions, reinforcement count, confidence	Vector top-5 behind a hard intent gate
RAG	Externally-ingested documents, chunked and embedded	Vector top-5, triple-gated
Memory slots	Seven server-enforced persistent slots (persona, preferences, project context, guidance, pending items, . . .)	Prepended verbatim to every prompt
Context clusters	Conversation-scoped topical groupings with unit-norm centroids	Restrict episodic search to relevant clusters

The **Codex** is the store most specific to conversation. Its central design lever is a three-bucket controlled relation vocabulary—*property* relations overwrite, *multi-valued* relations accumulate, and *single-valued* relations auto-expire the prior edge of the same (source, relation) pair—with generic relations such as `is` and `has` deliberately absent so the extractor is forced to be specific. Setting `valid_until` rather than deleting is what makes the graph temporal: an edge with `valid_until IS NULL` is currently true, and a non-NULL value means historically superseded but still auditable. Appendix E.2 details the write path and entity resolution.

Decay is access-weighted. Three decay workers apply per-cycle multipliers of roughly 5%, 2%, and 1% per day for unaccessed, accessed, and creative-tagged turns respectively, with a *creative floor* clamping decay score at 0.3 for narrative turns regardless of age—long-form fiction should not be summarised away for being old. Turns below 0.1 are archived and below 0.05 moved to cold storage. Codex edges decay similarly and are demoted from active to pending below a strength threshold; procedural patterns deactivate after 180 days with fewer than three reinforcements. Crucially, decay is not one-directional: every retrieval increments a fragment's access count and restores 0.15 to its decay score, so **retrieval is a partial reversal of forgetting** and what the system used yesterday is cheaper to reach today.

4.3 Hybrid Retrieval Orchestrator

The orchestrator is the core of pre-flight and the component this study measures most directly.

Retrieval legs. Six legs are defined: **BM25** over a Postgres tsvector of raw and summary text, filtered on decay score and archival status; **vector** with decay weighting, scoring $(1 - (\text{embedding} \Leftrightarrow \text{query})) \cdot \text{decay\_score}$, which is what distinguishes this leg from pure semantic search—a highly similar but decayed turn is down-ranked; **Codex graph traversal**; **procedural** behind a hard intent gate; **RAG**, triple-gated and global rather than conversation-scoped; and **batch summaries**, conversation-scoped. Section 8.1 reports which of these actually contributed during the evaluation, and why.

Dynamic leg weighting. Base weights are {bm25 0.8, vector 1.0, codex 0.5, procedural 0.2, rag 1.0}. Five intent profiles override them—Factual_Retrieval boosts vector and demotes codex; Generation, Ideation, and Open_Exploration do the reverse—and two cumulative topic overrides then apply. This is the mechanism by which inferred intent conditions retrieval, and it is the least-explored lever in the system.

Reciprocal Rank Fusion combines the legs:

$$\text{score}_{\text{RRF}}(f) = \sum_{\ell \in \text{legs}} \frac{\alpha_\ell}{k + \text{rank}_\ell(f)}, \quad k = 60, \tag{1}$$

where α_ℓ is the blended leg weight and $\text{rank}_\ell(f)$ starts at 1. Fragments are deduplicated *during* fusion: the first occurrence is registered and later occurrences add to its score, so agreement across legs is rewarded. Because RRF consumes ranks rather than scores, heterogeneous legs need no score normalisation to be combined—which is precisely why it can rescue a lexical leg whose raw scores are on an incomparable scale (Section 6.3).

Post-fusion curation applies five sequential transforms, and it is these—not the number of stores—that produce ICE's fragment reduction. Additive bonuses reward keyword overlap and length and penalise very short fragments; a recency bonus favours the most recent decile but is *skipped* for creative turns, because recent meta-discussion is noise for narrative work. Session diversification caps foreign conversations at three fragments each while leaving the active conversation uncapped. Deduplication hashes fragment text. A two-phase token budget then guarantees leg diversity—the best fragment of each leg is admitted first if it fits—before greedily filling the remainder. Finally the strengthening step writes back the access-count and decay-score restoration described above.

Cluster-scoped retrieval restricts episodic search to the most relevant topical clusters, falling back to global search when no cluster clears a threshold.

Dynamic token budget. A total context budget of 23,000 tokens less an overhead reserve leaves 21,200 available. A length-based recent-window fraction, reduced by token density and shifted by intent and topic, splits this between recent turns and retrieval, and a *growth cap* bracketed by turn count prevents long conversations from over-allocating. Leftover budget is intentionally left unused: this is the lever that keeps the retrieved set compact and, under extreme per-turn density, averts the context overflow that sinks an unbudgeted baseline (Section 6.2). On ordinary turns ICE's total token count is comparable to the baseline's—the curation shows up as fewer *fragments*, not fewer tokens. A configurable subclass of the orchestrator exposes flags that switch legs and post-processing steps on and off without redefining classification or fusion; Section 6.3's buildup uses that mechanism.

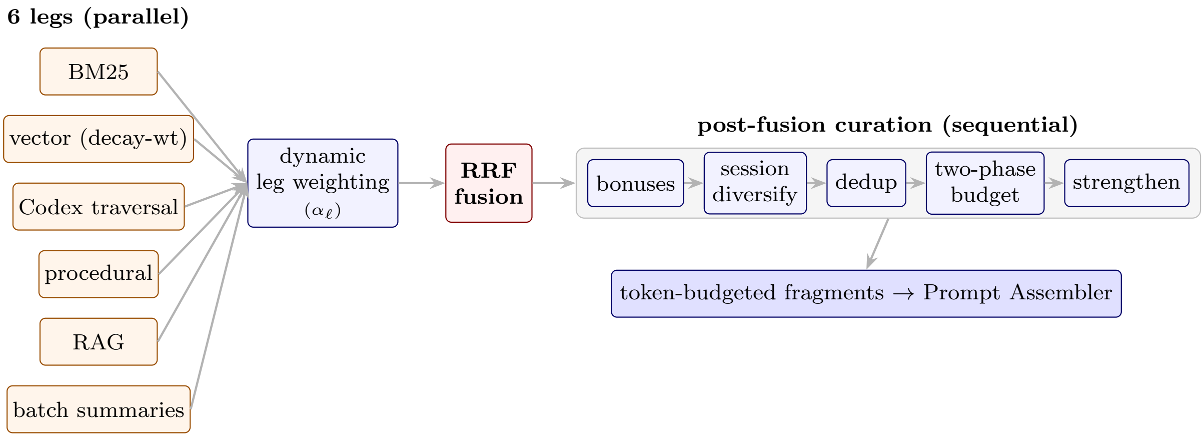

Figure 2: Retrieval orchestrator. The legs are heterogeneous in retrieval semantics; RRF is what allows them to be combined without score normalisation; the two-phase budget guarantees leg diversity under budget pressure. Section 8.1 reports which legs contributed during evaluation.

4.4 Prompt Assembly

The assembler orders messages for stable-prefix reuse: a fixed system message with an inline block rendering the active memory slots; recent turns under dynamic per-turn word caps; a single user message carrying the retrieved context; a short assistant acknowledgement acting as a boundary marker so the model treats the final message as the live question rather than another history turn; and the question itself. A second budget pass at the API layer enforces a per-model word ceiling, popping procedural and episodic fragments first and preserving document and graph fragments because they are hardest to recompute. Appendix E.4 gives the full ordering and caps.

This assembler is part of what the evaluation compares. The baseline receives retrieved chunks and the question; ICE receives fused, curated fragments *plus* slots, a recent window, and structural boundaries. That is a deliberate full-system-versus-standard-RAG comparison, stated explicitly in Section 5.3 so it is not mistaken for an accident.

5 Evaluation Protocol

5.1 Why a New Protocol Was Needed

Our first evaluation attempt measured retrieval *precision*: did the orchestrator retrieve the fragment whose timestamp the ground-truth file named? It failed for two compounding reasons. The first we call the *stateless pronoun problem*: probes were extracted from their conversations and tested in a vacuum, so a question like "you mentioned the current script saves nothing" reached the orchestrator stripped of every surrounding turn, and the system returned fragments that were mathematically correct in isolation and functionally useless. The second was an *incompetent judge*: a 7B grading model scored 0 for fragments a human immediately recognised as the exact session being asked about. Manual re-scoring of a subset produced 31% precision against the automated 13.87%, and the gap was the judge, not the retrieval.

Two conclusions followed, and they define LSREP. Evaluating retrieval intermediates is the wrong level of abstraction—a fragment that is irrelevant at turn 100 may be critical at turn 800, and a fragment differing from the reference may be equally valid—so the system must answer real questions and the *answers* must be judged. And if memory accumulation is the object of study, the same questions must be asked at multiple points in time, not once at the end.

5.2 LSREP

LSREP freezes a historical conversation at a temporal slice T_n, reconstructs the entire database and memory-lifecycle state exactly as it existed at that moment, injects evaluation probes, and evaluates the answers with an independent judge.

The general procedure: for each conversation, determine the turn count L and select checkpoints; wipe the target database and inject turns $T_1 \dots T_{n-1}$ sequentially into episodic memory, preserving relative or real timestamps; trigger the background workers synchronously, so decay computes initial scores, the graph extractor builds entities and edges, the procedural extractor scans for behavioural motifs, and the clustering worker aggregates related turns; then administer the probes and judge the answers. The protocol is system-agnostic: it requires only that the system under test expose a chat endpoint and that its persistent state be reconstructible from a turn sequence.

We ran two instantiations. **Experiment 1 is a pilot** whose role was to expose measurement flaws and immature subsystems; it is reported in Appendix B, which also tabulates the two instantiations side by side (table 10), and it is referenced here only where it explains a design change. **Experiment 2 is the substantive evaluation**, and **Experiment 3** is a cumulative ablation explaining why the mature system behaves as it does.

5.3 Experiment 2 Environment

System under test. Experiment 2 evaluated the mature system: the classifier backbone upgraded to Qwen3-Embedding-0.6B with the head retrained from scratch; a trained micro-NER model replacing the pilot's regex entity tagger, so the graph extractor grounds on real entity detection (~95% recall); the conservative routing override added; the static budget and uncapped sliding window replaced by the unified dynamic token budget of Section 4.3; and cluster-scoped retrieval introduced.

Protocol instantiation. Rather than many conversations at isolated checkpoints, Experiment 2 follows four long-horizon conversations across their entire lifespan. Three choices differ sharply from the pilot. *(1) Continuous replay with preserved state*: each conversation is replayed chronologically into one fresh deployment, and memory stored at an early checkpoint remains available at every later one; a simulated memory lifecycle—decay, edge decay, procedural decay, reflection, clustering, consolidation, sentinel monitoring—runs *between* checkpoints, so retrieval reflects aged, consolidated memory rather than fresh storage. *(2) Automatic, leg-aware probe generation*: probes are generated at each checkpoint from a sampled history window (first 5 turns, last 10 before the checkpoint, 15 random from between) using only information available up to that checkpoint, explicitly targeting specific legs, and each requires a unique temporal anchor to prevent ambiguity across hundreds of turns. These are combined with 72 hand-written probes inherited from the pilot. *(3) Incremental evaluation*: every probe whose origin is at or before a checkpoint is re-evaluated at that checkpoint, so a probe generated early is scored repeatedly as history accumulates—which is what makes longitudinal memory growth directly observable.

Evolving ground truth. The pilot's single frozen reference is replaced by a three-stage temporal-refinement pipeline: *forensic regeneration* of the origin-checkpoint answer under a stricter, evidence-oriented prompt that records temporal provenance; *temporal anchoring*, attaching to each probe an anchor identifying the fact or event it tracks; and *forward propagation*, updating each answer at every later checkpoint when new content contradicts, updates, or extends it, with anchor-preservation checks guarding against silent drift to a superficially similar later event. Each probe therefore carries a temporally evolving reference document rather than a static answer—necessary because otherwise a system that correctly *updated* a fact would score worse than one repeating an outdated one.

Scoring. Experiment 2 adds temporal-aware scoring (an answer loses points for reporting a superseded fact even if it was once correct) and inference credit (a specific, evidence-consistent detail not explicitly in the reference is rewarded rather than flagged as fabrication).

Human verification, not score alteration. Two pieces of human involvement need stating precisely, because "manually verified" can be misread as "scores were adjusted by hand." They were not; the human role was verification and correction of the judge's own labels, not substitution of human judgement for the judge's. First, the 72 hand-written probes inherited from the pilot were re-evaluated by hand against both their absolute scores and their tournament rankings, so a human-authored, human-checked slice sits inside the benchmark alongside the automatically generated and automatically judged probes—not merely the model's own self-consistency. Second, every hallucination flag across all 1,211 probes was manually audited. This was necessary because the automated judge was systematically too harsh across *all four* conditions—generalist and MoE, ICE and vector alike—whenever an answer contained a specific, true, directly relevant detail that was simply absent from the necessarily condensed ground-truth dossier; the judge tended to flag such details as unsupported rather than recognise them as correct supplementary information. The audit corrected these false positives wherever a detail was verifiably true against the source conversation, applying the same standard to every condition; it did not remove flags for genuine fabrications, and we deliberately did not over-correct by crediting speculative or unverifiable additions. This audit covers Experiment 2 only; Experiment 3 uses the automated judge's hallucination labels unmodified, for the bias reasons in Section 8.

Judge. Both experiments use the same independent judge, `gemma-4-12B-AWQ` served on SGLang with a 150,000-token context at temperature 0.0 and deterministic decoding, never used for retrieval, generation, probe generation, or ground-truth construction. The extraction rules—strict neutrality, verbatim anchor preservation, temporal dominance, third-person attribution—are shared; Experiment 2 applies them through the evolving-dossier pipeline rather than a single pass.

Conditions. A four-condition matrix: **Vector RAG (generalist)**—single-leg pgvector similarity, top-30, no decay weighting, classification, or fusion, on a generalist model; **Vector RAG (MoE)**—same retrieval, expert-routed; **Full ICE (generalist)**—all legs enabled, RRF fusion, classification, decay weighting, dynamic budget, post-fusion curation; and **Full ICE (MoE)**. Both use the same embedder and database. The baseline is a standard single-leg vector-RAG—top-30 similar turns assembled with the question—while the ICE conditions run the full system, fused retrieval *plus* ICE's prompt assembler. The comparison is therefore *full memory system versus standard vector-RAG*, which is what a deployment decision actually looks like: ICE's curation deliberately governs what context is assembled, not merely which fragments are retrieved.

A note on conservatism. Because history is *replayed* rather than lived, the reinforcement that normally strengthens frequently-referenced memories during use is largely absent—the primary reinforcement signal becomes the evaluation probes themselves. A concept referenced dozens of times in real use and one mentioned once therefore look more alike under replay than they would in deployment. Experiment 2's numbers should be read as a conservative lower bound on deployed performance, not a ceiling.

5.4 Metrics

No single metric captures memory quality, which is why we report several and read them together. Absolute *score* conceals the cost at which quality was achieved; *SPF* and *TUR* expose that cost, which is the axis on which a curated memory system must distinguish itself from a firehose. *Win rate* is the noise-robust relative signal, sidestepping the absolute-scale drift that makes raw 1–5 judgements hard to compare across hundreds of probes, and it often separates systems that tie on mean score. *Hallucination* and *fragment count* guard the two ways a system can look good on score and fail in practice—fabrication, and context bloat. *Recency delta* is the longitudinal signal unique to this setting: whether answers *improve* as memory accumulates, which a static-corpus baseline structurally cannot exhibit.

Where confidence intervals are reported they are 95% percentile-bootstrap intervals over 10,000 resamples of probes with replacement, computed on the same per-probe records that produce the point estimates and including the published missing-score imputation; paired score differences are computed within-probe before resampling. The scripts that produce them are released with the harness.

6 Results

The pilot (Experiment 1) is reported in Appendix B. Its short version: honestly measured, the immature system was *worse* than the vector baseline on every metric—4.04 against 4.06 on score, 47% more tokens, higher hallucination—and a token-accounting audit found the reported context size had omitted the recent-turn sliding window, which invalidated any efficiency claim once corrected. That result drove the mature design and is the reason this paper reports the pilot as diagnostic rather than as evidence.

6.1 Mature System Evaluation

table 3 reports the four conditions on the three-conversation view (Dataset C excluded; 1,057 probes). Per-conversation, score-distribution, and temporal-quality breakdowns are in Appendix D.

Table 3: Experiment 2, global comparison, Dataset C excluded (1,057 probes across 3 conversations). On the conversations where both systems can generate usable answers, ICE ties the vector baseline on mean score, injects 32% fewer fragments, and wins 30.6% of head-to-head comparisons against 21.2%.

Condition	Score	Tokens	Frags	SPF	TUR	Win %	Hall. %
Vector RAG generalist	4.25 ± 1.09	21,025	30.0	0.14	0.20	21.2	20.5
Vector RAG MoE	4.24 ± 1.04	21,025	30.0	0.14	0.20	19.2	17.4
Full ICE generalist	4.26 ± 1.08	22,411	20.4	0.21	0.19	**30.6**	19.6
Full ICE MoE	**4.28** ± 0.99	22,411	20.4	0.21	0.19	29.0	19.9

The mean-score difference between ICE generalist and vector generalist is +0.01 (4.26 vs. 4.25). A paired per-probe bootstrap—both conditions answer the same probes, so the difference is taken within-probe before resampling—places it at +0.00 with a 95% CI of $[-0.07, +0.07]$: a tightly estimated statistical tie rather than an ambiguous one. The secondary metrics carry the interesting content.

Efficiency. ICE injects 32.0% fewer fragments (20.4 vs. 30.0) at the same quality, giving an SPF of 0.21 against 0.14—a 50% improvement in quality per fragment. It uses 6.6% *more* tokens (22,411 vs. 21,025). That overage is not fragment bloat: it is prompt structure the baseline never receives. Every ICE prompt carries a fixed instruction preamble, a persistent-context header, a retrieved-context header and a boundary acknowledgment turn—about 195 tokens of static scaffolding—plus any active memory slots verbatim and a window of recent conversation turns, while the baseline is served only its retrieved fragments and the question. The comparison is therefore full-system-versus-standard-RAG rather than fragment-versus-fragment, and the accounting is deliberately unflattering to us: ICE spends its extra tokens on structure and recency while retrieving a third fewer fragments. We state it plainly because it is the opposite of the claim such systems usually make: the win is in *how many things* the model must reconcile, not in raw token count. Whether that scaffolding earns its cost is *untested*. We never ran ICE's retrieval behind the baseline's bare prompt, so we cannot say whether stripping the preamble and the recent-turns window would have been cheaper at equal quality, cheaper at worse quality, or neither: the tokens and the answer quality move together in our design and we did not separate them. This is a controlled comparison the study omits, not a result it reports, and it is the first arm we add in the follow-up.

Preference. The head-to-head win rate is the cleanest signal and it is statistically robust: ICE generalist wins 30.6% of tournament appearances (95% CI $[27.9, 33.4]$) against the baseline's 21.2% ($[18.8, 23.7]$), and the intervals do not overlap. When the two systems disagree, ICE is more often preferred. Hallucination rates are close (19.6% vs. 20.5%).

Fragment–score correlation shows a qualitative difference rather than a quantitative one. ICE generalist has $r = +0.193$ (MoE $+0.246$); vector generalist has $r = -0.015$ (MoE -0.030). For ICE, retrieving more fragments goes with better answers; for the baseline, retrieving more goes with worse. Same metric, opposite sign—a system whose retrieved fragments correlate negatively with answer quality is, in a deep sense, a system that does not know what it is doing.

Longitudinal behaviour. fig. 3 plots mean probe score against turn-index bin. All four conditions track closely through early and middle bins—the tie is visible as four overlapping curves. The separation appears at the deepest bin (turn 1,100 of Dataset B): both vector conditions drop sharply, to 3.80 and 3.87, while both ICE conditions hold at 4.18 and 4.14. The baseline's retrieval cannot reach across 1,000+ turns of history, and the gap opens precisely where that reach matters.

Token efficiency is not a constant, and the pooled average hides a crossover. The 6.6% figure above is an average over conditions that differ sharply, and two mechanisms push against each other. Per-turn *density* favours ICE: the baseline retrieves a fixed top-30, so its injected tokens scale with how large each turn is, without bound. Conversation *length* favours the baseline: ICE's growth cap deliberately widens the budget as turns accumulate ($2{,}000 + 150n$ below 30 turns, then $5{,}000 + 100(n - 30)$, then $10{,}000 + 30(n - 100)$), while top-30 is indifferent to how long a conversation has run. table 4 contrasts the first and last quartile of each conversation, paired per probe and bootstrapped. Within every conversation the baseline is effectively flat and only ICE moves, so the trend is a property of the budget rather than of which conversation is being read.

The clearest case is Dataset D, where the effect reverses sign inside a single conversation: ICE injects 2,309 fewer tokens than the baseline over the first quartile and 4,994 *more* over the last, both intervals excluding zero, against a baseline that barely moves. Dataset A shows the same erosion without a reversal, and Dataset C shows density overwhelming the effect entirely—there the baseline sits near 87,000 tokens and ICE is cheaper on every probe at every position. The practical reading is that a per-query budget buys the most when turns are dense and least when a conversation is long and sparse. What it does *not* show is that the extra spend is wasted. Paired per-probe score differences move the other way: from the first quartile to the last they improve in all four conversations—$+0.19$ to $+0.20$ on Dataset A, -0.23 to $+0.18$ on B, $+3.18$ to $+3.42$ on C, and -0.07 to $+0.16$ on D—so the conversations where ICE's relative cost rises are also the ones where its relative quality rises, consistent with the long-horizon divergence in fig. 3. We resist the causal reading: the budget was never varied independently of the retrieval it funds, so these are correlated consequences of accumulated memory, not a measured return on tokens. What it does license is narrower—a cap keyed to *turn count* spends on a schedule rather than on evidence. Replacing fill-to-cap with a marginal-relevance stopping rule would let the same budget be justified per query rather than per position, and it is the same fix the ablation's dynamic-budget regression pointed at.

Routing. The global MoE-versus-generalist score delta is $+0.01$ for ICE and -0.02 for vector; the effect is too small to claim in either direction. MoE gives a 3.1% hallucination reduction under vector and a 0.3% increase under ICE, and no token savings. The router ran on every probe and moved nothing—a genuine null, and we return to its significance in Section 8.1.

Memory maturity and reliability. Simulated decay days at the final checkpoint were 93 for Dataset B, 24 for A, 27 for C, 20 for D. Gating failures—probes where the classifier blocked retrieval that was needed—fell from 22 in the pilot to 2 here, the single largest reliability

Table 4: Token cost by position within each conversation: first quartile of probes versus last. Δ is the paired per-probe difference (ICE minus baseline; negative means ICE is cheaper) with a 95% bootstrap interval, and "cheaper" is the share of probes on which ICE injected fewer tokens. The baseline is close to flat within each conversation; ICE's cost rises with position in all four.

Dataset	Position	Paired Δ	95% CI	ICE cheaper
A (Creative, 290 turns)	first quartile	−8,034	[−8,650, −7,451]	100%
	last quartile	−2,284	[−2,766, −1,800]	91%
B (Long-form, 1,119 turns)	first quartile	+3,766	[+3,133, +4,394]	18%
	last quartile	+1,288	[+476, +2,075]	35%
C (Technical, 325 turns)	first quartile	−70,731	[−79,067, −63,734]	100%
	last quartile	−68,854	[−88,105, −53,696]	100%
D (Academic, 251 turns)	first quartile	−2,309	[−3,112, −1,521]	69%
	last quartile	+4,994	[+4,362, +5,609]	5%

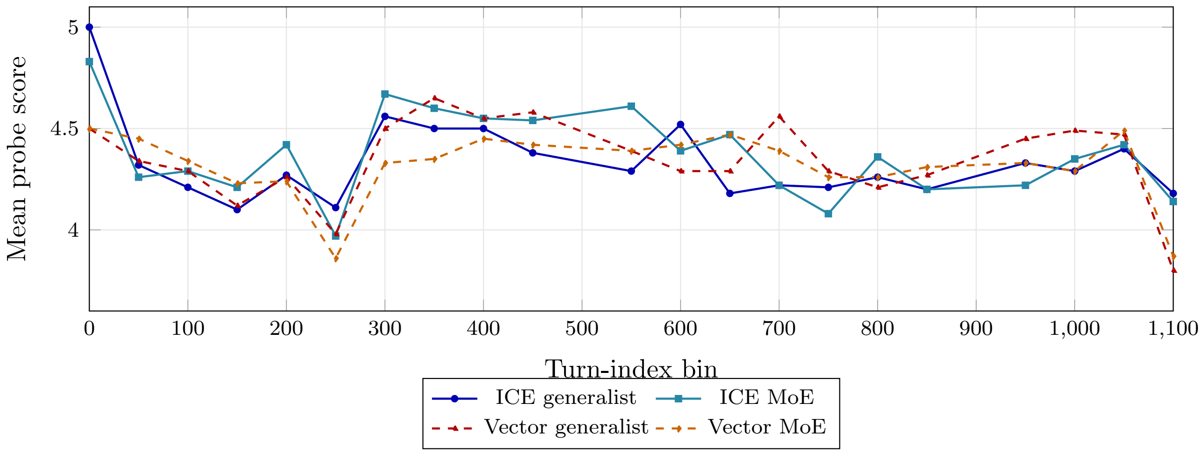

Figure 3: Longitudinal mean score by turn-index bin (1,057 probes, Dataset C excluded; real per-bin data). The four conditions track closely mid-conversation, but at the final turn-1,100 bin—the deep long-horizon regime of Dataset B—both vector conditions drop sharply while both ICE conditions hold.

improvement in the mature deployment.

Human-verified slice. The 72 hand-written probes were scored entirely by hand, both an absolute 1–5 score and a blind, anonymised tournament ranking, giving a human-authored, human-judged slice inside the benchmark (table 5). Here the picture is far less equivocal than the automated tie: both ICE conditions score around 4.3 against the baselines' 3.1, and the paired ICE-minus-baseline difference is +1.21 (95% CI $[+0.81, +1.61]$), decisively excluding zero. Two readings are consistent with this and we lean on neither. The optimistic one: the automated judge—shown in Section 8 to be systematically harsh—*understates* ICE's advantage, so the headline tie is a conservative lower bound. The cautious one: these 72 are the harder, more memory-dependent, author-written questions on which stored context matters most, and a single scorer, even blinded per probe, cannot be assumed unbiased. The slice's dependable value is narrower but real: it is an independent check that the automated tie is not concealing an ICE *deficit*. The within-ICE tournament asymmetry—the MoE variant taking 69.4% of first places despite a marginally lower mean—is not stable at this sample size and should not be read as an MoE effect.

Table 5: Human-verified slice (72 hand-scored probes, independent of the automated judge). Paired ICE-versus-vector is +1.21 (95% CI [+0.81, +1.61]), in contrast to the automated tie over all 1,057 probes—consistent with the automated judge being conservative. Single scorer; blind per-probe tournament ranking.

Condition	Score	95% CI	Win %
Vector RAG (generalist)	3.10	[2.75, 3.44]	5.6
Vector RAG (MoE)	3.22	[2.88, 3.54]	8.3
Full ICE (generalist)	4.31	[4.11, 4.49]	16.7
Full ICE (MoE)	4.24	[4.01, 4.43]	**69.4**

6.2 Stress Test: Context Overflow Under Extreme Density

Dataset C contains architecture documents that frequently exceed 8,000 tokens in a single turn. The vector baseline, which always retrieves the top-30 fragments with no per-query budget, injects a mean of 88,061 tokens—failed cases reach 100,505—overwhelming the model's context window. Of 154 probes, 145 fail with a score of 1, a 94.2% failure rate. The retrieval subsystem did find relevant information; the generation model could not process the assembled context and produced empty or degenerate responses. ICE, with its dynamic budget capping retrieval, holds a mean of 19,953 tokens and a mean score of 4.33.

Table 6: Stress test on Dataset C (154 probes). The vector baseline collapses because it has no token-budget cap. ICE survives with a mean score of 4.33 and 77% fewer tokens injected. The baseline's 0% hallucination rate is an artefact: 94.2% of its responses are complete failures, so there is no answer to evaluate.

Condition	Score	Tok.	Frags	SPF	TUR	Hall %	Win %	Overflow %
Vector RAG generalist	1.23	88,061	30	0.04	0.01	0.0*	1.9	**94.2**
Vector RAG MoE	1.10	88,061	30	0.04	0.01	0.0*	4.5	97.4
Full ICE generalist	**4.33**	19,953	10	**0.43**	**0.22**	41.4	43.5	3.9
Full ICE MoE	4.03	19,953	10	0.40	0.20	40.8	**50.0**	16.2

The score distributions are far apart. ICE generalist: 3.9% at score 1, 5.2% at 2, 11.7% at 3, 12.3% at 4, 66.9% at 5. The baseline: 94.2% at score 1, nothing at 2–4, 5.8% at 5. ICE's SPF of 0.43 is more than ten times the baseline's 0.04.

The 41.4% hallucination rate for ICE here deserves a careful note, because read naively it looks like a defect. ICE is attempting a difficult question and sometimes gets it wrong. The baseline is not attempting at all, so it never hallucinates; a 0% rate here is not evidence of reliability but evidence of absence. The rate reflects task difficulty on a dataset where the system has far more material to work with, not a failure mode unique to ICE.

Information density as a benchmark dimension. Turn count alone is an insufficient measure of difficulty. Dataset B is the greatest challenge in *temporal horizon*; Dataset C is a different challenge entirely, *extreme information density.* The benchmark surfaced two distinct stress dimensions, and the second was not designed—it emerged when the first few probes returned errors.

What the headline numbers look like with Dataset C included. table 3 deliberately excludes Dataset C, so the reported tie reflects conditions under which both systems can produce an answer; a 94.2% failure rate would otherwise dominate any aggregate without saying anything about answer *quality.* So that this choice is not mistaken for cherry-picking, table 7 reports the same four conditions over *all* 1,211 probes.

On this all-data view the paired per-probe difference is +0.40 (95% CI [+0.31, +0.49]), decisively excluding zero; on the stress test alone it is +3.10 ([+2.85, +3.33]). This is a stronger result than the headline tie suggests, and we want to be explicit about why we did not lead with it. Excluding Dataset C from the primary comparison is the more conservative choice, not

Table 7: Experiment 2 over all four conversations, Dataset C included (1,211 probes). Including the conversation where the baseline collapses does not merely narrow the gap—it *reverses* it: the baseline falls to 3.87 while ICE rises slightly to 4.27. The win-rate and efficiency gaps both widen relative to table 3.

Condition	Score	Tokens	Frags	SPF	Win %	Hall. %
Vector RAG generalist	3.87 ± 1.47	29,550	30.0	0.13	18.7	20.2
Vector RAG MoE	3.84 ± 1.45	29,550	30.0	0.13	17.3	15.7
Full ICE generalist	**4.27** ± 1.09	22,099	19.1	**0.22**	**32.3**	22.3
Full ICE MoE	4.25 ± 1.03	22,099	19.1	0.22	31.7	22.5

the more flattering one: it isolates "does ICE cost anything when both systems can answer" from "does ICE survive when the baseline cannot," so a reader can evaluate each claim on its own evidence rather than one number smuggling in the other. Reporting only the all-data table would invite the fair objection that a single catastrophic-failure conversation does all the work of the headline; reporting only the excluded-C table would omit a real finding. table 6 exists so the stress result is visible and auditable on its own, and table 7 exists so a reader can see exactly how much the exclusion was doing. We consider both necessary and neither sufficient alone.

6.3 Feature Ablation

To understand which mechanisms matter, we ran a cumulative feature buildup on Dataset B (67 probes, fully mature memory state, single pass), constructing the system incrementally from a bare-vector baseline with each step adding exactly one feature and preserving all previous ones. The judge here is a smaller 14B model, the only one available on the dedicated inference server without interrupting the main evaluation, so absolute scores sit below Experiment 2's; relative deltas are the object of interest. table 8 gives the CI-robust steps; the full fifteen-row table, the recency breakdown, and the waterfall chart are in Appendix C.

Table 8: The ablation steps that clear a paired within-probe bootstrap 95% CI (10,000 resamples, 67 probes scored under every condition), with the endpoints for context. Every other step's interval spans zero; the full table is table 12. The single-leg vector baseline is a reference point, not a buildup step.

Step	Score	Paired Δ	95% CI	
bare_vector	3.27	—	—	
+BM25	2.52	−0.74	[−1.14, −0.36]	excludes 0
+RRF	3.36	+0.82	[+0.39, +1.24]	excludes 0
nine further steps, all spanning zero (Appendix C)				
full_ice	3.38	−0.03	[−0.29, +0.23]	
vector_baseline (ref.)	3.42	—	—	

Adding an unfused lexical leg is harmful (−0.74, [−1.14, −0.36]). Unfused lexical matches flood the context and actively hurt answer quality—naive multi-leg retrieval is worse than single-leg retrieval, which is the opposite of the intuition that motivates most multi-signal designs.

Rank fusion is corrective (+0.82, [+0.39, +1.24]). Adding RRF on top recovers the damage. Its role is precisely that: recovery. Neither RRF-on-vector nor the fully built configuration is statistically distinguishable from the single-leg baseline on answer quality—both contrasts span zero, consistent with the Experiment 2 tie. RRF is the mechanism that makes multi-signal retrieval *safe*, not a lever that lifts quality above a strong baseline. We state this deliberately because the earlier draft of this work claimed RRF was the single most impactful component; the paired intervals do not support that, and the corrected claim is the smaller one.

The remaining steps must be read with care, and Section 8.1 explains why: several toggled features were not functioning or not varying at the snapshot, so their near-zero deltas measure nothing about the mechanism. The genuinely *measured* neutrals—features that ran and moved nothing—are cluster restriction and session diversification ($\Delta \leq 0.01$). One design lever is implicated negatively: the dynamic budget's step is -0.11, because its fill-to-cap policy allocates more tokens to longer conversations and over-injects on Dataset B (21.2 fragments against 11.3 at the pre-budget step). The ceiling should be a guardrail; filling to it is the wrong policy, and Section 9 describes the marginal-relevance stopping rule that replaces it.

6.4 Which Mechanisms Carried the Result

Because Section 8.1 reports that part of the retrieval design did not contribute, it is worth stating positively—and only with evidence—which mechanisms did. This is not a list of everything ICE implements. It is the subset for which this study contains direct evidence, and it is a longer list than the retrieval legs alone, because ICE is not only its retrieval legs.

The dynamic token budget. The entire stress-test result is this one mechanism. An unbudgeted retriever with an identical embedder and database over the identical conversation fails on 94.2% of probes; adding a per-query budget takes the same corpus to a mean of 4.33. Paired difference $+3.10$ $[+2.85, +3.33]$. Of everything in the system, this is the single most consequential component, and it is also the least sophisticated.

Rank fusion. The corrective effect above, CI-robust in both directions (-0.74 then $+0.82$). It is the mechanism that lets heterogeneous legs be combined at all.

Decay-weighted episodic retrieval. BM25 and the decay-weighted vector leg together supply 62.3% of injected fragments—the substrate everything else curates. Decay weighting is what separates this leg from plain semantic search: a highly similar but stale turn is demoted rather than surfaced.

The Codex knowledge graph. Live throughout Experiment 2 and contributing 3.3% of fragments. That figure is a lower bound by construction, and it reflects a representation choice rather than a failure. The evaluated implementation collapsed an entire graph traversal—every visited entity and its context payload—into a *single* concatenated fragment per probe, whereas the episodic legs emit each turn as its own fragment. A traversal carrying a dozen entities therefore counted as one fragment against dozens of individually counted episodic ones, and could occupy at most one slot under budget enforcement regardless of relevance. Entity detection was not the problem: the trained micro-NER model was loaded and running at roughly 95% recall. The Codex in this build is **structurally under-weighted, not broken**, and its 3.3% should be read as a floor on its content share, not a measurement of its value.

Post-fusion curation. Bonuses, session diversification, deduplication, and cluster scoping are what turn 30 candidate fragments into 20.4 injected ones at equal quality. The 32% fragment reduction and the 50% SPF improvement are their aggregate effect, and the positive fragment–score correlation is the evidence that what survives curation carries signal.

Decay and retrieval reinforcement. Every retrieval incremented access counts and restored decay scores throughout the replay, so the ranking at checkpoint n depends on what was retrieved at checkpoints $1 \ldots n-1$. This shaped every ranking in the study. We cannot size its individual contribution—it was never toggled—but it was running, and the simulated decay horizons (93 days on Dataset B) are what the longitudinal curves are measured through.

Classification and the retrieval decision. Gating failures fell from 22 to 2 between the pilot and the mature system. That is a reliability result about the decision to retrieve at all, and it is the precondition for everything downstream; Section 8 is candid that a conservative override, not the classifier alone, deserves part of the credit.

Prompt assembly. Memory slots, the recent-turns window, and stable-prefix ordering are part of what the comparison measures. We flag this as a scope statement rather than a confound: the study compares a full memory system against standard vector-RAG, which is the comparison a deployment decision actually faces, and managing recent context is something a memory system is supposed to do.

Mixture-of-experts routing. It ran on every probe, was measured, and came out neutral (+0.01 for ICE, −0.02 for vector). We list it here rather than among the untested components because the distinction matters: this is a *genuine null*. The mechanism executed and moved nothing, which is a finding about the mechanism. Section 8 explains why we think the router, not the idea of routing, is what failed.

7 Discussion

7.1 Memory Is Curation, Not Collection

ICE's advantage comes not from retrieving more—it retrieves less—but from retrieving better. The token budget, rank fusion, and the post-fusion transforms are the mechanisms that measurably filter noise here; decay, cluster scoping, and session diversification are designed to the same end and were neutral on the one dataset we could ablate. A memory system that remembers everything is a memory system that remembers nothing useful. The fragment–score correlation is the cleanest empirical statement of the principle: for ICE more fragments go with better answers ($r = +0.19$), for the baseline more fragments go with worse ($r = -0.02$).

The stress test sharpens it further. The situation in which curation matters most is not the one where retrieval is hard; it is the one where retrieval is *easy and abundant*. Dataset C's documents are highly relevant to its probes. The baseline finds them, injects them all, and fails—not because it retrieved the wrong things but because it retrieved too many right things. No improvement in ranking would have saved it. Only a decision to stop would have.

7.2 What the Ablation Can and Cannot Establish

The buildup cleanly establishes two things: an unfused lexical leg is harmful, and rank fusion recovers it. It cannot adjudicate most of the features it toggles, because several were defective, disabled, or structurally under-weighted at the snapshot—a flat delta from a component that never ran is uninformative, not a domain-specific null. The one design lever it does implicate is the budget's fill-to-cap policy, which points not at "more retrieval is bad" but at the need for a marginal-relevance stopping rule.

Beyond that, the honest reading is that the load-bearing mechanisms in this study were the unglamorous ones—fusion, the budget, and post-fusion curation—while the more ambitious stores await a build in which they are fully exercised. The classification-driven leg weighting, which conditions retrieval on inferred intent, remains the least-explored lever and the most likely source of future gains.

7.3 Generalisation

The budget finding should generalise to any system injecting retrieved context into a fixed-window model: the failure mode is a property of the interface, not of our retriever. The fusion finding should generalise to any multi-leg retrieval system, and it carries a design warning—adding a leg is not free, and a leg added without fusion can be worse than no leg at all.

The methodological finding is the one we would most like to see travel. **A component must be verified genuinely live before a null can be attributed to it.** Ablation is the standard instrument for attributing credit in multi-component systems, and it silently produces the same output—a delta near zero—for "this does not help" and for "this did not run." Those two readings license opposite engineering decisions. The remedy is cheap and we would now consider it mandatory: assert non-empty per-component contribution inside the harness, and publish a per-component contribution table beside the results. We did neither, discovered the gap by post-hoc audit, and report the consequences in the next section.

8 Limitations

We are deliberately specific about what this study does and does not show. Several of these limitations directly motivate Section 9.

8.1 What the Benchmark Did and Did Not Exercise

A component-level audit of the evaluated build (git tag `v2-paper-eval`) separates three states that systems evaluations routinely collapse into one. We report them separately because the distinction changes what a reader is entitled to conclude, and because collapsing them—as an earlier draft of this work did—is both harsher and less accurate than the evidence supports. Appendix A gives the component-by-component table.

(A) Genuinely defective: one component. The *procedural* retrieval leg bound its query embedding as a plain array rather than a typed pgvector value. Under the deployed driver every call therefore raised a type error on `vector <=> double precision[]`, rolled back the transaction, and returned an empty list. The leg was 100% dead across every experiment, and the hard intent gate in front of it meant nothing surfaced the failure. This is a bug, plainly. It is fixed post-snapshot, and no result in this paper depends on procedural memory.

(B) Never exercised by this benchmark: several. These are limitations of the *test*, not findings about the features. For each, the correct reading is absence of evidence, not evidence of absence, and we claim nothing about them in either direction.

- *Document retrieval.* No document was ever ingested into the store during these experiments, so the leg was never given input. (It shared the binding defect above, but even repaired it would have had nothing to retrieve.)
- *Batch summarisation and cold storage.* These exist for turns that have decayed past the point of individual retrieval. No turn in the replay decayed that far—the longest simulated horizon is 93 days—so the replay never reached the regime these mechanisms are built for. The implementation was correct; the store was empty.
- *Temporal retrieval.* All 1,211 probes ask for *current* truth. Not one asks how a fact changed over time, which is the query class a temporally-versioned graph exists to serve. The versioning ran and was maintained correctly; nothing in the benchmark interrogated it.
- *Cross-conversation retrieval.* Every probe is scoped to a single conversation, so within-conversation memory is demonstrated and cross-conversation aggregation is simply untested.
- *Query rewriting (HyDE).* Disabled outright in Experiment 2. In Experiment 3's build it ran on every long-term-memory query in *every* arm—gated by the context-reliance label rather than by the ablation flag—so it was active in the bare-vector step too and the +HyDE step toggled nothing. It was never isolated in either experiment.

(C) Measured and neutral: one. Mixture-of-experts routing ran on every probe under both retrieval conditions and produced a global delta of +0.01 (ICE) and −0.02 (vector). This is a genuine null, and we distinguish it sharply from (B): the mechanism executed, was measured, and moved nothing.

What follows. The reported results—the quality tie, the 32% fragment reduction, the context-overflow survival—were produced by the mechanisms enumerated in Section 6.4: two working episodic legs, a live but under-weighted knowledge graph, rank fusion, decay, post-fusion curation, prompt assembly, and the token budget. If anything this *sharpens* the central thesis, since the system matched a strong baseline without the more elaborate stores contributing much. But we state the boundary precisely rather than either direction of overstatement. **A component that never ran is not a component that failed**, and this study licenses no conclusion about the ceiling of the knowledge graph, document memory, temporal retrieval, or query rewriting. Equally, it does not license the reading that most of the system failed: one component was broken, several were untested, one was measured neutral, and the rest were working and contributing.

The transferable lesson is procedural. Had the harness asserted a non-empty fragment count per leg and emitted a per-leg contribution table with the results, the defective bind would have surfaced on the first probe of the first run instead of in a post-hoc audit months later. We now regard that assertion as a required part of evaluating any multi-component retrieval system, and Section 9 builds it into the follow-up study.

8.2 Single-User Evaluation

All benchmark conversations were authored by one person. They span creative writing, technical planning, academic planning, and long-horizon worldbuilding, so they exercise a diverse set of conversation *types*—but they reflect the habits, vocabulary, and interaction style of a single individual, and they demonstrate nothing about a diverse *population of users*. This is the study's most consequential limitation and it is structural: it cannot be repaired by re-analysis, only by a different corpus. The follow-up study addresses it with a releasable synthetic corpus and an external anchor benchmark.

One property of the corpus cuts the other way. It is demanding rather than convenient: turn lengths vary by orders of magnitude, one conversation runs past 1,100 turns, and another's turns routinely exceed 8,000 tokens—the case on which the baseline fails outright. Performance here is thus a lower bound with respect to density and horizon. We resist the tempting extension: a hard corpus does not license transfer to other users, since one author also fixes topic coverage, what counts as a correct answer, and who wrote the reference answers.

8.3 Replay Is Not Live Use

The evaluation reconstructs history by replaying existing conversations and probing the resulting memory state. This permits controlled experimentation over thousands of turns but cannot reproduce the feedback loops of live use, where retrieval events continuously shape future memory through reinforcement, decay adjustment, bookmarking, and user behaviour. During replay, future turns already exist and cannot be influenced by retrieved output.

The specific consequence is on reinforcement. ICE strengthens frequently accessed memories and lets unused ones decay; under replay, reinforcement comes almost entirely from benchmark probes rather than natural use. In real conversation a user might return to a character's motivations every few sessions, reinforcing that memory repeatedly; here it is reinforced only when a probe targets it. The benchmark therefore likely *underestimates* the benefit of reinforcement-driven maturation, and the positive longitudinal curves should be read as a lower bound.

8.4 The Classifier Is Safe Rather Than Smart

Between the pilot and Experiment 2 the classifier was retrained on a substantially better embedding backbone, cutting gating failures from 22 to 2. But a second change landed at the same time: a conservative routing override applied *after* classification, upgrading Zero_Shot to Long_Term_Memory whenever confidence fell below threshold or the conversation exceeded a turn count.

One concrete failure mode motivated it—a *public-entity trap*, in which the classifier associated well-known proper nouns with Zero_Shot because the model already "knows" who they are, even when the query was scoped to the user's own long-running conversation about that entity. A question naming a famous character from Dataset A's fan-fiction was liable to be classified as needing no retrieval at all. The override is effective but crude: it cannot distinguish a public entity discussed in a genuinely stateless way from one that is the subject of extensive private elaboration.

The override is a deliberate safety net—false negatives are fatal, false positives merely expensive—and the pilot's evidence supported it. But it means the system is no longer purely intent-gated in the way the architecture claims, because the heuristic frequently forces retrieval over the classifier's judgement. ICE is therefore "safe" rather than "smart" here, and we cannot claim the gating is driven by the learned classifier when a rule is actively compensating for its uncertainty.

8.5 Codex Extraction Reliability

The knowledge graph is the most architecturally ambitious component and, in this implementation, the most fragile. Its 3.3% fragment share and +0.03 ablation step reflect several simultaneous engineering handicaps rather than a theoretical ceiling: extraction chunks were oversized for the 4B extraction model, so entities introduced mid-passage were dropped and distant clauses conflated; edge confidence and strength are nearly inert at retrieval time, with no reinforcement loop raising frequently referenced facts; and resolution is entity-centric, so a user who establishes "The Obsidian Citadel" and later asks about "the main fortress" gives the resolution step no anchor to match. Appendix E.3 expands each. The marginal contribution should be read as a floor: that it contributed anything despite oversized chunks, a single-fragment retrieval representation, and no reinforcement suggests a properly implemented graph could become one of the stronger components.

8.6 Mixture-of-Experts Routing

The neutral routing result has three plausible causes, all deficiencies of this router rather than of routing. Routing is driven by hardcoded topic and intent mappings rather than learned or empirical per-model performance, so there is no evidence a model tagged for software questions actually outperforms a generalist on them. The router ignores classifier confidence, treating 0.55 identically to 0.95. And it ignores context-reliance, routing a self-contained query identically to a context-heavy one. An operational cost compounds these: selecting an unloaded model forces a 5–15 s model swap outside ICE's control. The infrastructure is conceptually underdeveloped and operationally expensive, which is why we read the null as a verdict on this implementation.

8.7 Judge Limitations and the Ablation Audit Gap

Evaluation relied primarily on an automated judge. During analysis, multiple cases were identified where correct, true information was flagged as hallucinated because it was absent from the *condensed* ground-truth summary rather than from the conversation—a systematic over-harshness affecting all four conditions equally, not a bias against any one system. Every hallucination flag across all 1,211 Experiment 2 probes was therefore individually audited against the source conversation and corrected where the flagged detail was verifiably true; flags on genuine fabrications were left in place and no credit was given for speculative additions. This is label correction, not score substitution: the judge's own criteria were applied more carefully, not replaced by human opinion.

Experiment 3 was *not* audited this way. Because it evaluates fifteen closely related variants under identical conditions, manual adjustment risked introducing subjective bias across comparisons meant to isolate a single change at a time. Its absolute hallucination values should therefore be treated as potentially inflated by the same false positives; relative differences between conditions remain valid, but its absolute numbers should not be compared directly with Experiment 2's audited ones.

8.8 Further Scope Limitations

Probe generation bias. Most probes came from a constrained model-assisted pipeline, and generated questions may differ from the anaphoric, pragmatically ambiguous questions real users ask. The 72 hand-authored probes partially mitigate this, but the benchmark remains weighted toward generated probes. **Domain coverage.** The four domains omit collaborative software development, multi-user conversation, customer support, multilingual dialogue, and workplace communication, where multi-party dynamics may change the picture. **Decay horizon.** The longest simulated horizon is 93 days; year-scale stability and saturation remain open. **Prompt-cache benefit.** Stable-prefix ordering is the correct design but delivers little in practice—the recent-turn window changes every request, model swaps wipe the cache, and only the system message and slots (roughly 10–15% of the prompt) are reliably reusable. **Hardware.** All experiments ran on a single 24GB consumer GPU; the ablation used a 14B judge rather than the 26B generalist to fit fifteen conditions in reasonable time, so its absolute scores are systematically lower. **No commercial comparison.** Cloud assistants' memory features cannot be run locally with controlled retrieval, so the vector-RAG baseline is the strongest fair comparison constructible on identical hardware.

9 The Programme This Study Opens

The limitations above define a research programme, and we describe it in two registers: what has already been built in the released system since the evaluated snapshot, and what the next study will measure. None of it affects the numbers reported here, all of which reflect tag `v2-paper-eval`.

9.1 What the Released System Already Changes

Each gap this evaluation exposed has an implemented or specified fix in the released code, post-dating the snapshot.

Extraction. The passive triplet collector is replaced by a reconciliation-based pipeline: deterministic entity grounding produces a confirmed entity list, relationship mapping is constrained to those entities—removing the degenerate self-referential triplets—and every proposed triplet is checked against the graph, so a contradiction opens a bounded reconciliation loop, with ambiguous supersessions escalated to a review queue rather than blind-written.

Density. The stress-test collapse motivated retrieval at the 512-token chunk level plus a raw/summary/abstract representation hierarchy, so a single 8,000-token turn can no longer dominate the budget, and the ceiling now derives from the routed model's actual context window rather than a constant.

The defective bind. The procedural and document legs' typed-vector binding is repaired, so both execute rather than rolling back.

Maintenance. The fixed worker set is orchestrated by a maintenance agent with an explicit tool surface, reconciling graph state during idle GPU time, with the review queue preserved as the human-in-the-loop boundary.

The untested axes. A temporal-retrieval layer answers as-of, range, and evolution queries—the axis no probe in this study exercised. A conversation-import path ingests exported chat logs through the very replay pipeline used here for evaluation, reconstructing episodic, graph, procedural, and cluster state rather than a searchable archive.

We list these as context for the next study, not as results. None has been measured under LSREP, and this paper claims nothing on their behalf.

9.2 What the Next Study Measures

A second study is underway that re-runs this protocol against the repaired system, and it is designed against exactly the weaknesses reported here. Confidence intervals are computed from the first probe rather than added post hoc. Per-component contribution assertions run inside the harness, so a leg that returns nothing fails the run instead of surviving to a later audit. Memory-deletion rules and expected effect sizes are pre-registered before the first probe executes, so a null is interpretable. A matched-prompt baseline—the baseline given the same recent-turns window—isolates retrieval from prompt assembly. A releasable synthetic corpus with a planted-fact ledger removes the can't-share-the-data limitation, with an external benchmark as anchor. And a new temporal probe class, scored on era-correctness and evolution narration, exercises the versioning this study left untouched.

What remains genuinely open is not on that list: a learned router conditioned on confidence, context-reliance, and empirical per-model performance; an opt-in continual-learning loop folding user feedback into classifier fine-tuning; and an explainability layer exposing per-fragment retrieval attribution over an editable graph.

The role of the present paper in that programme is specific. It establishes the protocol, and it establishes the baseline the follow-up will be measured against—including the parts of the baseline that are unflattering. That is what makes the candour an asset rather than an admission: a study that reports which of its own components were defective, which were never exercised, and which carried the result gives the next study something to beat, and gives a reader a way to check whether it did.

10 Conclusion

We have presented ICE, a local-first conversational memory system combining episodic memory, a temporally-versioned knowledge graph, procedural and document stores, intent-driven retrieval routing, and a dynamic per-query token budget; and LSREP, a longitudinal protocol that measures not retrieval quality at a single instant but memory accumulation, retention, and belief revision across months of real conversational history.

Evaluated under LSREP over 1,985 turns and 1,211 probes, ICE matches a strong vector-RAG baseline on answer quality while injecting 32% fewer fragments, wins a substantially larger share of blind head-to-head comparisons, and survives a context-density stress test where the unbudgeted baseline fails on 94.2% of probes. A cumulative ablation shows that unfused multi-signal retrieval is harmful and that rank fusion is corrective. A component-level audit separates one genuinely defective component, several the benchmark never exercised, and the mechanisms that carried the result—and we argue that keeping those three categories distinct is a methodological requirement, not a courtesy, because ablation produces the same near-zero delta for "does not help" and "did not run."

As context windows grow, the temptation is to stuff more context in. Our results suggest the opposite. Less-but-better context matches a firehose on raw quality while beating it on efficiency, on head-to-head preference, and on survival under density—and the one place the two designs diverge catastrophically is the place where retrieval was *easy*, where the baseline found many relevant documents and drowned in them. The future of conversational memory is not bigger windows; it is a system that knows when to stop retrieving. We release the implementation, the protocol, and the ablation harness so that others can test that claim against corpora we do not have.

Reproducibility and Ethics

Reproducibility. The full source, the evaluation harness, and the experimental configuration are released. All reported results reflect the system at the evaluation snapshot, preserved as git tag `v2-paper-eval`; the tag preserves the evaluated *code*, while companion artefacts written afterwards live on the main branch, and the implementation has continued to evolve since. Every number in Section 6 derives from shareable JSON reports, and the bootstrap scripts that produce the confidence intervals are released alongside them. Raw judge outputs and the probe sets are not released, because they contain personal conversational content.

One distinction deserves stating plainly: the *protocol* is fully reproducible, but the *data* is not manufacturable. The four corpora are real conversations accumulated through genuine use over months to years—Dataset B alone represents roughly a year of interaction—and their evaluation value derives precisely from that authenticity. The entity evolution, contradictions, superseded decisions, and long-range dependencies are organic, not planted. An equivalent corpus cannot be synthesised on demand, and this is a general property of longitudinal memory evaluation rather than a limitation specific to this study. Reproducing the evaluation therefore means bringing one's own long-form conversational history—most platforms support chat-log export, which the replay harness ingests—and the protocol, harness, judge configuration, and metrics transfer unchanged to any such corpus. The follow-up study's synthetic corpus is intended to complement this, not to replace it: a shareable benchmark and an authentic one measure different things.

Ethics. All four datasets were created by the sole author, who is the only data subject. No third-party conversational data was used and no institutional review was applicable. The system stores the user's data on the user's own hardware; nothing leaves the local machine. The architecture includes a memory-slot system in which high-stakes steering slots require explicit user approval before automated updates are applied, while low-stakes bookkeeping is autonomous—a deliberate human-in-the-loop boundary rather than a performance limitation.

Societal implications. By making persistent structured memory viable on a single consumer GPU, this architecture lowers the barrier to capable personal AI that does not depend on a metered cloud API. The same persistence is dual-use: a system that faithfully models a user's evolving beliefs, preferences, and history could be turned toward profiling or manipulation as readily as toward assistance, particularly if the store were moved off the user's control. We raise this not because the present work enables it—the system is single-user, local, inspectable, and fully exportable and deletable by construction—but because the properties that make it benign are precisely the ones a harmful variant would strip away. We regard keeping those user-sovereignty properties non-negotiable as part of the contribution rather than incidental to it.

References

Prateek Chhikara, Abhinav Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, abs/2504.19413, 2025. URL https://arxiv.org/abs/2504.19413.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '09)*, pages 758–759, New York, NY, USA, 2009. Association for Computing Machinery. doi: 10.1145/1571941.1572114.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A Graph RAG approach to query-focused summarization. Technical report, Microsoft Research, 2024. URL https://arxiv.org/abs/2404.16130.

Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1762–1777, Toronto, Canada, 2023. Association for Computational Linguistics. URL https://aclanthology.org/2023.acl-long.99/.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. REALM: Retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *PMLR*, pages 3929–3938, Virtual, 2020. PMLR. URL https://proceedings.mlr.press/v119/guu20a.html.

Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 874–880, Online, 2021. Association for Computational Linguistics. URL https://aclanthology.org/2021.eacl-main.74/.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pages 9459–9474, Red Hook, NY, USA, 2020. Curran Associates Inc. URL https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, abs/2310.08560, 2023. URL https://arxiv.org/abs/2310.08560.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019.

Chufan Shi, Cheng Yang, Xinyu Zhu, Jiahao Wang, Taiqiang Wu, Siheng Li, Deng Cai, Yujiu Yang, and Yu Meng. Unchosen experts can contribute too: Unleashing MoE models' power by self-contrast. *arXiv preprint arXiv:2405.14507*, abs/2405.14507, 2024a. URL https://arxiv.org/abs/2405.14507.

Weijia Shi, Sewon Min, Michihiro Yasunaga, Minjoon Seo, Rich James, Mike Lewis, Luke Zettlemoyer, and Wen-tau Yih. REPLUG: Retrieval-augmented black-box language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 8371–8384, Mexico City, Mexico, 2024b. Association for Computational Linguistics. URL https://aclanthology.org/2024.naacl-long.463/.

Jiashuo Sun, Chengjin Xu, Lumingyuan Tang, Saizhuo Wang, Chen Lin, Yeyun Gong, Lionel M. Ni, Heung-Yeung Shum, and Jian Guo. Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, Vienna, Austria, 2024. OpenReview.net. URL https://openreview.net/forum?id=nnVO1PvbTv.

Yu Wang, Nedim Lipka, Ryan A. Rossi, Alexa Siu, Rui Zhang, and Tyler Derr. Knowledge graph prompting for multi-document question answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19206–19214, Washington, DC, USA, 2024. AAAI Press. URL https://arxiv.org/abs/2308.11730.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884*, abs/2401.15884, 2024. URL https://arxiv.org/abs/2401.15884.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19724–19731, Washington, DC, USA, 2024. AAAI Press. URL https://ojs.aaai.org/index.php/AAAI/article/view/29946.

A Component Fidelity Audit

table 9 is the component-by-component audit summarised in Section 8.1, grounded in the code at tag `v2-paper-eval` and in the frozen Experiment 2 result JSON. It is reproduced in full because the three-way classification—defective, never exercised, contributing—is only checkable if the per-component evidence is visible.

Two notes on reading this table. First, the "unknown" 34.5% of fragments in Experiment 2—fragments whose originating leg could not be reconstructed from the stored records—is a logging limitation, not a fourth category of component; those fragments came from the legs that were running. Second, the entity-detection question is worth isolating because it is easy to get backwards: the pilot (Appendix B) genuinely used a regex tagger and had effectively no usable graph, but Experiment 2 loaded a trained model at roughly 95% recall. The mature system's weak graph contribution is a chunk-size and representation problem, not a detection problem, and conflating the two would misdirect the repair.

B Experiment 1: The Pilot

System under test. The pilot evaluated an immature build with only the vector and BM25 legs meaningfully active. The classifier used a frozen MiniLM backbone; retrieval used static per-leg limits, a static 5,000-token global budget, and an uncapped 10-turn sliding window; the knowledge graph effectively did not exist—a regex tagger and an under-powered 1.5B extractor produced almost no usable edges—so the graph, procedural, and document legs contributed nothing and the system reduced to essentially the same vector-plus-BM25 retrieval as the baseline; cluster-scoped retrieval did not exist yet.

Protocol instantiation. Conversations shorter than 10 turns were excluded, reducing the corpus from 42 to 18. Split points were drawn from $[\max(10, 0.30L), 0.95L]$, three per conversation with a fixed seed; each split defined a historical block and a 10-turn future reference block. Probes were hand-written, 8–30 per conversation, and probes from earlier splits were re-asked at later ones. Ground truth used a single retrieval-assisted pass, frozen thereafter. Six conditions ran per probe.

The token-accounting audit. After the pilot completed, an audit found that the reported context size for full ICE included retrieved fragments but *omitted* the recent-turn sliding window incorporated during prompt assembly. Generated answers, scores, rankings, and hallucination labels were unaffected—the window was correctly supplied to the model at inference time; only the reported efficiency statistics were wrong. We replayed the sliding-window assembly for every checkpoint and added the omitted tokens. The correction did not change the qualitative findings, but it invalidated any claim of superior token efficiency: corrected, full ICE consumes *more* total tokens than the baseline at roughly equal quality. All pilot numbers reported here are corrected values.

The headline numbers were a defeat: 4.04 against 4.06, 47% more tokens, more hallucination, a worse TUR. The picture brightened only on the longitudinal axis—193 tracked question curves showed scores improving over time on repeated probes, demonstrating memory accumulation. Several subsystems were broken or immature: only three decay cycles had run; the graph effectively did not exist; 22 gating failures meant the classifier was actively blocking retrieval on probes that needed it; and the budget was static.

Lessons. A system that has not had time to consolidate memory is not a fair test of a memory architecture. Honest token counting is essential—the under-count would have made ICE look more efficient than it was. And a broken subsystem can drag down an entire system, which is itself a finding about the fragility of multi-component architectures and a direct ancestor of the fidelity audit in Appendix A.

Table 9: Component state at the evaluation snapshot. "Contributing" means the component executed and affected retrieved output; "never exercised" means the benchmark supplied no input that would engage it; "defective" means it executed and failed.

Component	**State**	**Evidence**	**Claimable as**
Vector leg (decay-weighted)	Contributing	Typed vector bind present; dominates leg contributions	Validated
BM25 leg	Contributing	tsvector leg, no embedding bind needed; with vector, 62.3% of fragments	Validated
RRF fusion	Contributing	$+0.82$ $[+0.39, +1.24]$ over a lexical and a dense leg	Validated, scoped to two legs
Dynamic token budget	Contributing	Stress test: 94.2% failure without, mean 4.33 with	Validated; strongest single result
Post-fusion curation	Contributing	32% fewer fragments at equal quality; SPF +50%	Validated in aggregate
Decay + reinforcement	Contributing	Simulated decay days per conversation (93/24/27/20); access-count and score restoration on every hit	Running; size not isolated
Cluster scoping	Contributing	Correct bind; ablation $\approx +0.01$	Measured neutral
Session diversify, dedup, bonuses	Contributing	Post-fusion transforms; keyword boost +0.12	Contributing
Codex graph	Contributing, under-weighted	3.3% of fragments, one concatenated fragment per probe; NER loaded and running at ~95% recall	Floor on content share; *not* an entity-detection failure
Classifier + gate	Contributing	Gating failures 22 → 2	Validated, with the override caveat
MoE routing	Measured neutral	Ran on every probe; $\Delta + 0.01$ / -0.02	Genuine null
Procedural leg	**Defective**	Untyped array bind ⇒ `vector <=> double precision[]` ⇒ rollback ⇒ `[]`; 0.0 fragments	Bug; no claim
Document (RAG) leg	Never exercised	No documents ingested; store empty; leg also shared the bind defect	No claim either way
Batch summaries	Never exercised	Bind correct; nothing decayed far enough to batch; store empty	No claim either way
Cold storage	Never exercised	Replay horizon never reached the archival regime	No claim either way
Temporal / timeline retrieval	Never exercised	All 1,211 probes ask for current truth	No claim either way
Cross-conversation retrieval	Never exercised	Every probe scoped to one conversation	No claim either way
HyDE rewrite	Never isolated	Exp 2: disabled. Exp 3: gated by context-reliance, so active in *all* arms including bare vector; the flag toggled nothing	No claim either way

Table 10: How the two LSREP instantiations differ. The shared skeleton—state reconstruction, synchronous worker trigger, output-level judging by the same independent judge—is described in Section 5.2.

Dimension	**Experiment 1 (pilot)**	**Experiment 2 (mature)**
System maturity	Vector + BM25 legs; MiniLM classifier; static budget; uncapped sliding window; graph non-functional	All legs enabled; Qwen3 classifier + routing override; dynamic budget; cluster-scoped retrieval
Corpus	18 conversations (isolated checkpoints)	4 long-horizon conversations (full-lifespan replay)
Split range	$[\max(10, 0.30L), 0.95L]$, 3 splits/conv	Length-adaptive checkpoints (8–20), even spacing with jitter
Probes	Hand-written (8–30/conv)	Auto-generated, leg-aware (147) plus 72 inherited hand-written
Ground truth	Single retrieval-assisted pass, *frozen*	Three-stage forensic regeneration + forward propagation, *evolving*
Checkpoint state	Independent per checkpoint	Preserved across checkpoints; lifecycle runs between them
Probe evaluation	Each probe once	Incremental: re-scored at every later checkpoint
Scoring	Static correctness	Temporal-aware + inference credit; hallucination flags audited for false positives
Conditions	Six (incl. sliding-window controls)	Four (controls dropped)

Table 11: Experiment 1 (immature system), 657 probes, corrected token counts. After correcting the sliding-window under-count, ICE used more tokens than the vector baseline, scored slightly lower, hallucinated more, and had a worse TUR. The system was not yet a fair test of the architecture.

Condition	**Score**	**Tokens**	**Hall. (%)**	**TUR**	**Win %**
Control baseline (generalist)	3.05 ± 1.61	3,998	74.1	0.76	11.3
Control MoE	3.01 ± 1.61	3,998	74.4	0.75	7.6
Vector RAG (generalist)	**4.06** ± 1.31	**5,847**	**64.7**	**0.69**	22.9
Vector RAG (MoE)	4.05 ± 1.32	5,847	66.7	0.69	15.6
Full ICE (generalist)	4.04 ± 1.33	8,586	69.6	0.47	22.0
Full ICE (MoE)	3.96 ± 1.33	8,586	72.3	0.46	20.6

C Full Ablation Results

Table 12: Cumulative feature addition on Dataset B (67 probes), full buildup. Paired within-probe bootstrap, 10,000 resamples. Only +BM25 and +RRF clear a 95% CI; every other step spans zero. Steps whose feature was defective, never exercised, or never isolated at the snapshot (+procedural, +batch summaries, +HyDE) are *not* interpretable as neutral findings, and +Codex is under-weighted by construction (Section 8.1). The vector baseline is a reference point, not a buildup step.

Step	Score	Step Δ	Paired Δ	95% CI	SPF	Tokens	Frags
bare_vector	3.27	—	—	—	0.23	14,873	14.0
+BM25	2.52	−0.75	−0.74	$[-1.14, -0.36]$	0.17	14,868	15.2
+RRF	3.36	+0.84	+0.82	$[+0.39, +1.24]$	0.29	14,870	11.7
+HyDE	3.39	+0.03	+0.03	$[-0.13, +0.21]$	0.29	14,870	11.6
+cluster_restrict	3.40	+0.01	+0.01	$[-0.16, +0.21]$	0.30	14,911	11.3
+session_diversify	3.40	+0.00	+0.00	$[-0.22, +0.22]$	0.30	14,911	11.2
+codex	3.43	+0.03	+0.03	$[-0.21, +0.25]$	0.30	14,912	11.3
+MERA	3.22	−0.21	−0.21	$[-0.43, +0.01]$	0.28	14,912	11.4
+procedural	3.39	+0.17	+0.16	$[-0.07, +0.40]$	0.30	14,912	11.4
+batch_summary	3.41	+0.02	−0.02	$[-0.24, +0.23]$	0.30	14,912	11.3
+dynamic_budget	3.30	−0.11	−0.08	$[-0.36, +0.18]$	0.16	24,489	21.2
+sliding_window	3.32	+0.02	−0.02	$[-0.24, +0.23]$	0.16	27,763	21.3
+keyword_boost	3.44	+0.12	+0.08	$[-0.22, +0.37]$	0.23	27,769	15.1
full_ice	3.38	−0.06	−0.03	$[-0.29, +0.23]$	0.22	27,768	15.0
vector_baseline (ref.)	3.42	—	—	—	0.11	16,373	30.0

Headline contrasts under the same paired bootstrap: the BM25 damage is -0.74 $[-1.12, -0.36]$ and the RRF rescue $+0.82$ $[+0.39, +1.23]$, both excluding zero; RRF-versus-bare is $+0.09$ $[-0.18, +0.36]$, full ICE versus bare is $+0.09$ $[-0.24, +0.42]$, and full ICE versus the single-leg baseline is -0.06 $[-0.40, +0.27]$, all spanning zero.

Two steps warrant individual comment. **MERA**, the enumeration fallback for category queries (-0.21, $[-0.43, +0.01]$, not significant), fires only when entity extraction returns nothing; since detection was working, it seldom triggered, so this delta rests on a small effective sample and is a downstream indicator of extraction gaps rather than an independent evaluation. **Codex** (+0.03) was live but emitted its entire traversal as one fragment, so a near-zero step delta is expected regardless of graph quality.

Table 13: Recency effect by origin split (fact age). Probes grouped by the turn index at which they were created. Recency Δ is mean(late) − mean(early). The fused configurations show positive recency—answers improve as the conversation grows—while the single-leg baseline is slightly negative.

Condition	Early (0–400)	Mid (400–800)	Late (800–1200)	Recency Δ
bare_vector	3.00	2.83	3.42	+0.42
+BM25	2.20	3.05	2.49	+0.29
+RRF	3.00	2.83	3.54	+0.54
+keyword_boost	2.90	3.28	3.60	+0.70
full_ice	3.10	3.25	3.47	+0.37
vector_baseline	3.50	3.15	3.47	−0.03

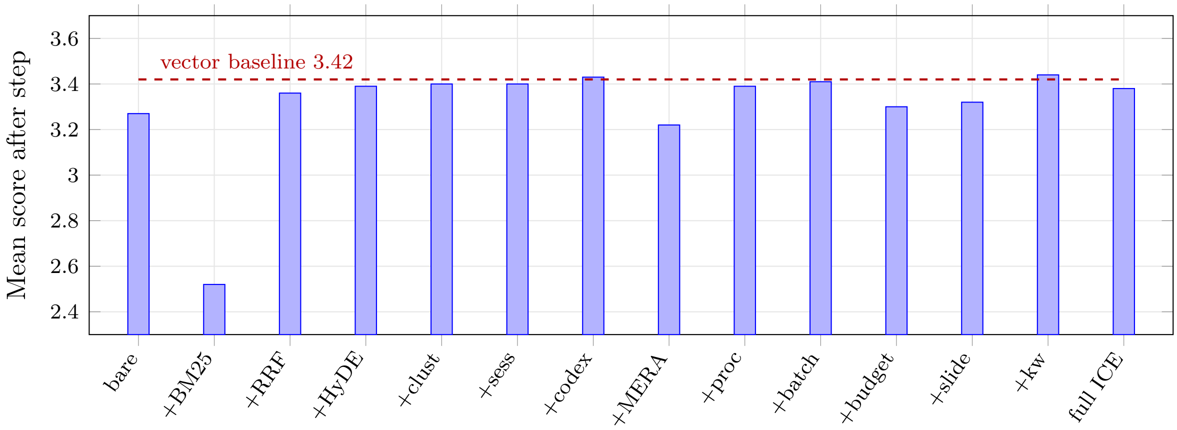

Figure 4: Feature ablation buildup (Dataset B, 67 probes). Each bar is the cumulative mean score after adding one feature to all previous ones. The only two movements clearing a paired bootstrap CI are the +BM25 drop and the +RRF recovery. The dashed line marks the single-leg vector baseline, which is not a buildup step; the full build finishes level with it.

D Experiment 2 Detailed Breakdowns

Table 14: Evaluation metrics. Each is computed per probe and aggregated across the benchmark.

Metric	Definition	What it measures
Score (1–5)	Mean absolute score from the judge	Answer quality
SPF	mean score/mean fragments	Retrieval efficiency
TUR	mean score/(mean tokens/1000)	Token efficiency
Win rate (%)	Blind tournament: all four conditions' answers for a probe are anonymised, shuffled, and ranked best-to-worst in a single pass, independent of absolute scoring; the win rate is the share of appearances placed first	Relative preference, free of judge order bias
Hallucination (%)	Share of probes where the judge detects information not present in the conversation	Faithfulness
Fragment count	Mean fragments injected per probe	Retrieval volume
Recency Δ	mean score(late bins) − mean score(early bins)	Longitudinal improvement

Table 15: Per-conversation breakdown (generalist conditions). ICE wins more head-to-head comparisons on every conversation. On Dataset A its win rate is more than double the baseline's.

Dataset	ICE score	Vec. score	ICE tok.	Vec. tok.	ICE win %	Vec. win %
A (Creative Writing)	3.63	3.50	19,434	24,360	**37.2**	18.0
B (Long-Form Creative)	4.28	4.33	24,108	21,257	32.8	22.9
D (Academic Planning)	**4.63**	4.57	20,103	18,076	20.4	18.8

Domain matters. On Dataset A (creative writing) ICE's win rate is more than double the baseline's. On Dataset B, the largest conversation at 1,119 turns, the baseline edges ahead on mean score (4.33 vs. 4.28) while ICE wins on win rate (32.8% vs. 22.9%) and uses 18% more tokens serving more detailed answers. On Dataset D ICE wins on both.

Table 16: Score distribution (1,057 probes, Dataset C excluded). The mean-score tie masks a structural difference: ICE-MoE has the lowest score-1 rate but shifts mass from 5 to 3–4, producing lower-variance output.

Score	**ICE gen (%)**	**ICE MoE (%)**	**Vec. gen (%)**	**Vec. MoE (%)**
1 (poor)	3.0	1.5	3.8	2.4
2	5.9	4.3	4.3	3.9
3 (acceptable)	13.3	16.6	14.8	19.5
4 (good)	18.1	19.7	17.2	16.3
5 (excellent)	59.7	58.0	60.0	58.0

Table 17: Temporal score quality (1,057 probes, Dataset C excluded). Buckets: Good (4–5), OK (3), Poor (1–2).

Condition	**Good (4–5) %**	**OK (3) %**	**Poor (1–2) %**
Vector RAG generalist	77.2	14.8	8.0
Vector RAG MoE	74.3	19.5	6.2
Full ICE generalist	**77.8**	13.3	8.9
Full ICE MoE	77.7	16.6	5.8

E Implementation Reference

This appendix collects operational detail supporting Section 4 but not required to follow the argument. A complete technical reference—the full training pipeline for the classifier and NER models, the controlled relation vocabulary in its entirety, GPU resource management, and the idempotency architecture—is the archived technical report in the repository, which describes the system at the tagged evaluation snapshot.

E.1 Classification Engine

The engine produces, per user turn, a set of topic tags, a set of intent tags, and one context-reliance label. These drive every downstream decision: which legs are weighted, how the budget is split, whether the wide-net fallback fires, and which model the router selects.

The learned head is deliberately tiny—a $384 \rightarrow 128 \rightarrow 25$ multi-layer perceptron with ReLU and 0.3 dropout. Its input is a single 384-dimensional embedding from a frozen `Qwen3-Embedding-0.6B` encoder; the shared linear head is sliced at inference into three blocks: 11 topic labels, 11 intent labels, and 3 context-reliance classes. Topic and intent decode multi-label (sigmoid, threshold 0.3, argmax fallback); context-reliance decodes single-label.

The rule-based pre-classifier computes five density signals in $[0, 1]$ from the raw prompt—code density, sentiment density, meta density (references to the model itself), noise density, and reference density (anaphoric terms)—and evaluates five rules in order, returning the first that fires or none, in which case the learned head runs. The reference rule is special: it returns empty topic and intent tags but forces Long_Term_Memory, and the orchestrator then runs the head only to obtain topic and intent. A two-tier threshold on that rule (0.2 for short conversations, 0.1 past ten turns) is the engine's primary long-conversation memory bias.

Override rules run in evaluation order after either path: memory immutability, so once Long_Term_Memory is set nothing may downgrade it; a topic rule promoting creative turns to Long_Term_Memory; and a rule promoting technical turns containing anaphora. A second, API-level bias lives outside the classifier: past ten turns, or below 0.95 confidence, a Zero_Shot label is upgraded before retrieval runs. Section 8 discusses what this costs.

Context-aware classification. When a conversation ID is supplied, the learned path queries the last three episodic turns for that conversation (max 500 words), preferring summaries and falling back to the first 150 words of raw text, and prepends that context to the prompt before embedding.

E.2 Codex Write Path and Retrieval

The graph spans four tables: entities (canonical name, aliases, properties JSONB, auto-regenerated context payload, embedding); edges (typed relations with strength, a confidence flag in {pending, active}, `valid_from`, and `valid_until`, where NULL means currently true); an append-only event log; and snapshots.

The triplet write path is the heart of temporal versioning. Entity resolution is two-stage—exact canonical-name match, then alias match, else create with a deterministic UUIDv5. The write branch then depends on the relation bucket: a property observation expires prior active edges of the same (source, relation) pair, writes a new edge, and overwrites the JSONB key; a reinforcement of an existing non-property edge increases strength and promotes from pending to active at strength ≥ 2; a new relation between the same pair expires the old one only if the old relation is not multi-valued. Every state change emits an event, and the entity's context payload is rebuilt from the property map plus the ten most recent active outgoing edges. The event log is compacted by a worker that snapshots any entity exceeding 100 uncompacted events—textbook event sourcing, with the live edge table as current state, the log as audit trail, and snapshots as bounded replay.

Micro-NER is a BIO tagger (a $384 \rightarrow 128 \rightarrow 64 \rightarrow 3$ MLP over per-token embeddings). When the trained model is unavailable the system falls back to a capitalisation regex minus a stoplist; at the Experiment 2 snapshot the trained model was present and loaded. **Vector fuzzy matching** embeds the extracted entity strings, scans entity embeddings, and takes the best above a cosine threshold of 0.85—the mechanism that resolves a slightly misspelled proper noun to its canonical entity.

The retrieval leg extracts prompt entities, resolves them by fuzzy matching, optionally restricts to a conversation scope, and performs a BFS traversal to depth 3 over edges where `valid_until IS NULL`, appending a labelled context payload per visited entity. **MERA**, the enumeration fallback for category queries, activates only when NER extracts no entities *and* the prompt contains both a category trigger and an enumeration hint; candidates are deduplicated and ranked by 30-day mention count.

E.3 Codex Limitations in Detail

The chunking paradox. Extraction used 6,000-token chunks with 200-token overlap, chosen to avoid splitting sentences and to preserve narrative context. A 4B-parameter model does not have the effective reasoning capacity to process 6,000 tokens at once; beyond roughly 1,000–1,500 tokens attention dilutes, losing entities mentioned mid-chunk while over-weighting the beginning and end. Two failure modes follow: *entity dropping*, where entities introduced mid-passage are never extracted, and *relationship hallucination*, where subject and object of distant clauses are confused, producing syntactically valid but meaningless triplets. The right chunk size for a 4B extractor is closer to 512–800 tokens.

Inert confidence and strength. Edges carry a confidence flag and a decaying numeric strength, but both are under-used at retrieval time. Traversal follows any edge with `valid_until IS NULL` regardless of either, and the only effect of an active confidence flag is a coarse 1.5× fragment-level score boost. There is no reinforcement loop strengthening edges on retrieval, so frequently referenced facts do not rise in the ranking and traversal treats weak edges as equal to strong ones.

Lexical versus semantic entity matching. Retrieval resolves entities by canonical name or alias plus fuzzy matching over the context-payload embedding rather than the entity name. A user who establishes "The Obsidian Citadel" and later asks "where is the main fortress located?" gives the resolution step no lexical or embedded anchor for "fortress," though a human reader resolves it immediately. This is a general limitation of entity-centric retrieval in free-form conversation: it assumes users refer to entities by canonical names or close aliases, which natural dialogue does not honour.

E.4 Prompt Assembly

The assembler emits a message list in a deliberately stable-prefix order to maximise cache reuse: (i) a fixed system message with an inline persistent-context block rendering each active memory slot; (ii) recent turns as alternating user/assistant pairs under dynamic per-turn word caps; (iii) a single user message headed with the retrieved context, optionally tagged with cluster names when a cluster scope is populated; (iv) a single assistant acknowledgement acting as a boundary marker, so the model treats the final user message as the live question rather than another history turn; and (v) the live question. Because the system message, slots, and most of the recent-turn prefix change slowly, most of the prefix key/value tensors are in principle reusable across consecutive requests—though Section 8 reports that the practical benefit is small.

A two-layer budget enforces final size: the orchestrator packs fragments to the retrieval token ceiling, then the API layer recomputes total words against a per-model ceiling (2,770 words for the deployed model) and iteratively pops procedural and episodic fragments until the budget is met, preserving document and graph fragments because they are the hardest to recompute and the most likely to carry the answer.

Memory slots are prepended verbatim to every prompt. Seven slot names are server-enforced: persona, user preferences, tool guidelines, project context, guidance, pending items, and session patterns. Slots are written through four paths: direct user update, batch initialisation, reflection-proposed but user-gated updates (high-stakes slots land in a review queue), and reflection-applied updates to pending items only.

Context clusters are conversation-scoped topical groupings. The centroid of member turn embeddings, renormalised to unit length after every change, is a 384-dimensional vector; membership is many-to-many. The clustering worker assigns each unassigned turn to the best existing cluster—combining embedding similarity, a bonus per shared entity, and tag overlap—or opens a new cluster when no candidate clears a 0.6 similarity threshold.

E.5 Background Workers and Infrastructure

All GPU-touching workers gate on a utilisation threshold polled from the driver and, in shared-model mode, additionally on a user-activity key with a short idle window. Retry countdowns are fixed per class. All workers are idempotent through a unique idempotency key, turning at-least-once delivery into effectively-once side effects.

The system is a single service exposing an OpenAI-compatible chat-completions endpoint plus routers for memory slots, user control, and the model registry, backed by PostgreSQL with pgvector—one 384-dimensional vector column per vector table, because the embedder is shared—and a worker fleet over a message broker. Configuration is a typed settings object read from the environment. Server-sent event types (`classified`, `retrieval`, `context_ready`, `generating`, `degraded`) drive an observability panel and downstream monitoring. Classifier fine-tuning writes a timestamped checkpoint but does not auto-promote it, so production promotion requires a deliberate swap.

Table 18: Background workers. The cluster is self-maintaining: no operator intervention is required for the memory lifecycle.

Worker	**Trigger**	**Function**
Post-flight evaluator	every turn	lossless detection, summary, dispatch extractors
Codex extractor	lossless turns	triplet extraction and versioned writes
Procedural extractor	every turn	pattern detection, reinforcement or insertion
Episodic decay	1.5 h	access-weighted decay, archive at 0.1, cold-store at 0.05
Codex decay	1.5 h	edge strength decay, demote below strength 0.3
Procedural decay	1.5 h	deactivation after 180 d with < 3 reinforcements
Clustering	30 min	assign unassigned turns, merge similar clusters
Reflection	2 h	session synthesis, pattern crystallisation, slot proposals
Batch summariser	2 h	50-turn batches, preservation-prompt summaries
Sentinel monitor	30 min	declarative rules: threshold, absence, and others
Fine-tune	weekly	retrain the classifier head on curated labels
Compaction	manual	snapshot entities with ≥ 100 uncompacted events
Drop zone	filesystem	ingest documents into the document store
Codex inject watcher	filesystem	ingest structured files into the graph with manual confidence